

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Project Scenario: Railway Reservation System

Name	Rohit Pujari
SRN	PES1UG25CS836
Section	K
Course	Software Engineering

1. Project Overview

This report documents the Agile backlog creation and sprint simulation carried out in Jira for the Railway Reservation System project. Functional requirements identified in Lab 1 were broken down into Epics and User Stories, prioritized, estimated using story points, and executed across simulated sprints.

1.1 Epics Summary

Epic 1: User Registration & Authentication

Enable passengers to create accounts, log in securely, and manage their profile so bookings can be tied to a verified identity.

Epic 2: Train Search & Availability

Allow passengers to search for trains between stations, view schedules, and check seat availability across classes.

Epic 3: Seat Booking & Payment

Enable passengers to select seats, enter passenger details, and complete secure payment to confirm a reservation.

Epic 4: Ticket Management & Cancellation

Let passengers view, modify, cancel bookings, and track refund or waitlist status after a reservation is made.

Epic 5: Admin & Train Schedule Management

Allow railway staff/admins to manage train schedules, seat inventory, and fares, and to monitor bookings across the system.

2. Epics and User Stories

Epic 1: User Registration & Authentication

Enable passengers to create accounts, log in securely, and manage their profile so bookings can be tied to a verified identity.

ID	User Story	As a / I want / So that	Priority	Points
1.1	Account Registration	As a a new passenger I want register using my name, email and phone number So that I can create an account to book train tickets	High	3
1.2	Login	As a registered user I want log in with my email and password So that I can access my bookings and account details	High	2
1.3	Password Recovery	As a user who forgot my password I want reset it via an OTP sent to my registered email/phone So that I can regain access to my account	Medium	3
1.4	Profile Management	As a logged-in user I want update my personal details and saved passengers So that my booking information stays accurate	Low	2

Epic 2: Train Search & Availability

Allow passengers to search for trains between stations, view schedules, and check seat availability across classes.

ID	User Story	As a / I want / So that	Priority	Points
2.1	Search Trains	As a passenger I want search trains by source, destination and date So that I can see all available trains for my journey	High	5
2.2	View Seat Availability	As a passenger I want view real-time seat availability by class (Sleeper, AC, General) So that I can decide which class to book	High	3
2.3	View Fare Details	As a passenger I want see the fare breakdown for each class before booking So that I can budget my trip accordingly	Medium	2
2.4	Filter & Sort Results	As a passenger I want filter trains by departure time, duration or price So that I can quickly find the most suitable train	Low	3

Epic 3: Seat Booking & Payment

Enable passengers to select seats, enter passenger details, and complete secure payment to confirm a reservation.

ID	User Story	As a / I want / So that	Priority	Points
3.1	Passenger Details Entry	As a passenger I want enter details (name, age, gender, berth preference) for each traveller So that the ticket is issued with correct passenger information	High	3
3.2	Seat/Berth Selection	As a passenger I want select or auto-assign my preferred berth So that I can travel comfortably as per my preference	Medium	5
3.3	Payment Processing	As a passenger I want pay securely using UPI, card or net banking So that my booking is confirmed instantly	High	8
3.4	Booking Confirmation	As a passenger I want receive an e-ticket and confirmation via email/SMS So that I have proof of my reservation	High	3

Epic 4: Ticket Management & Cancellation

Let passengers view, modify, cancel bookings, and track refund or waitlist status after a reservation is made.

ID	User Story	As a / I want / So that	Priority	Points
4.1	View Booking History	As a user I want view all my past and upcoming bookings in one place So that I can track my travel plans	Medium	2
4.2	Cancel Ticket	As a passenger I want cancel a booked ticket before departure So that I can get a refund if my plans change	High	3
4.3	Track PNR / Waitlist Status	As a waitlisted passenger I want check my PNR status and waitlist position So that I know whether my seat is confirmed	High	3
4.4	Refund Tracking	As a user who cancelled a ticket I want track the status of my refund So that I know when the amount will be credited back	Low	2

Epic 5: Admin & Train Schedule Management

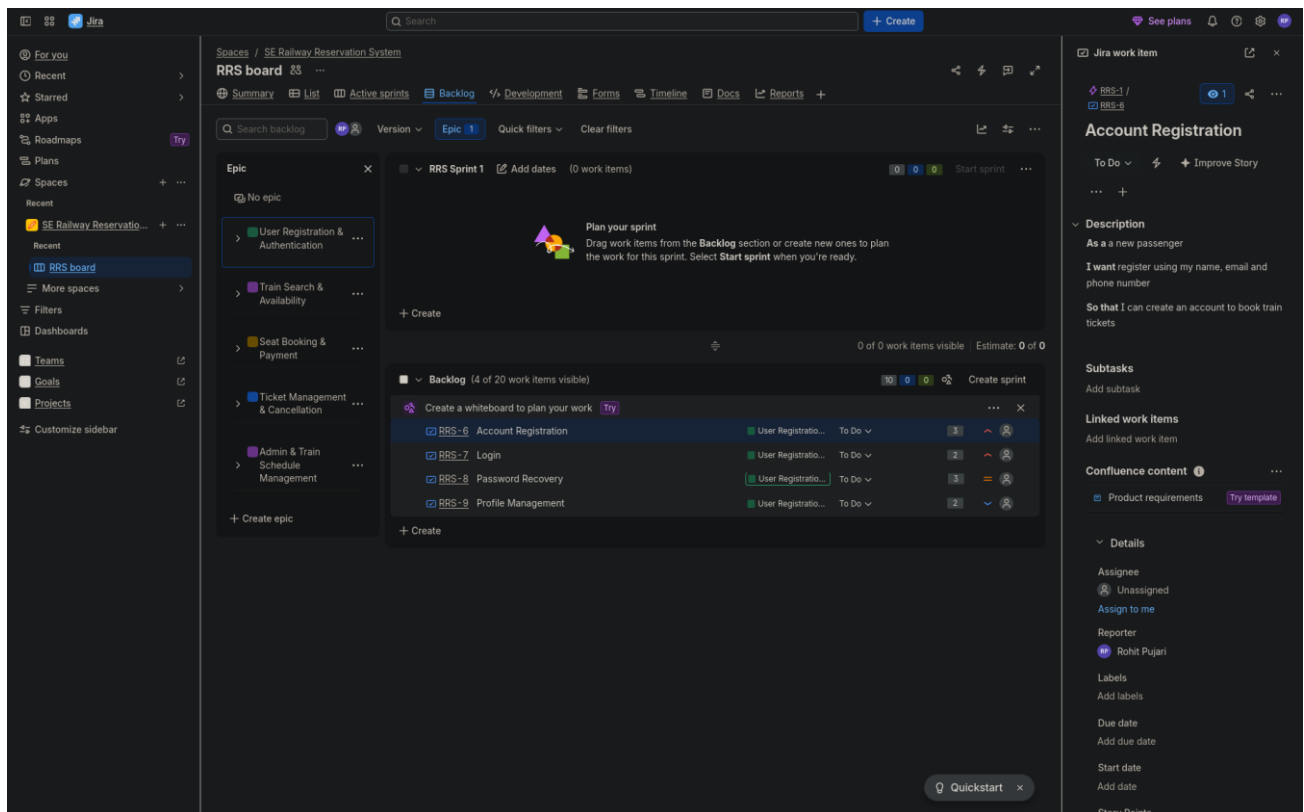
Allow railway staff/admins to manage train schedules, seat inventory, and fares, and to monitor bookings across the system

ID	User Story	As a / I want / So that	Priority	Points
5.1	Add/Update Train Schedule	As a railway admin I want add new trains and update their routes, timings and stops So that the search results always reflect accurate schedules	High	5
5.2	Manage Seat	As a railway admin	High	3

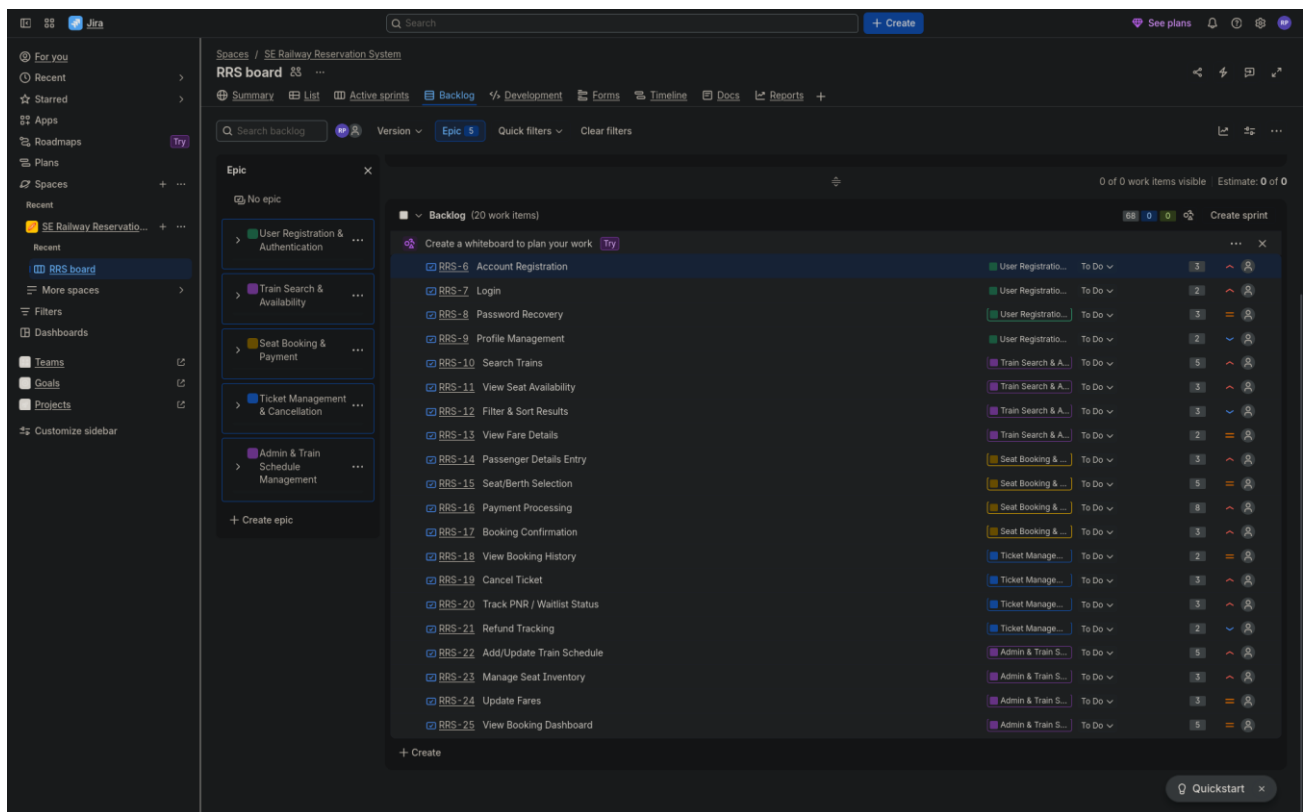
	Inventory	I want configure the number of seats per class for each train So that availability shown to passengers is accurate		
5.3	Update Fares	As a railway admin I want set and update fares per class and route So that passengers are charged the correct amount	Medium	3
5.4	View Booking Dashboard	As a railway admin I want view a dashboard of daily bookings, cancellations and occupancy So that I can monitor system usage and plan capacity	Medium	5

3. Jira Workspace Screenshots

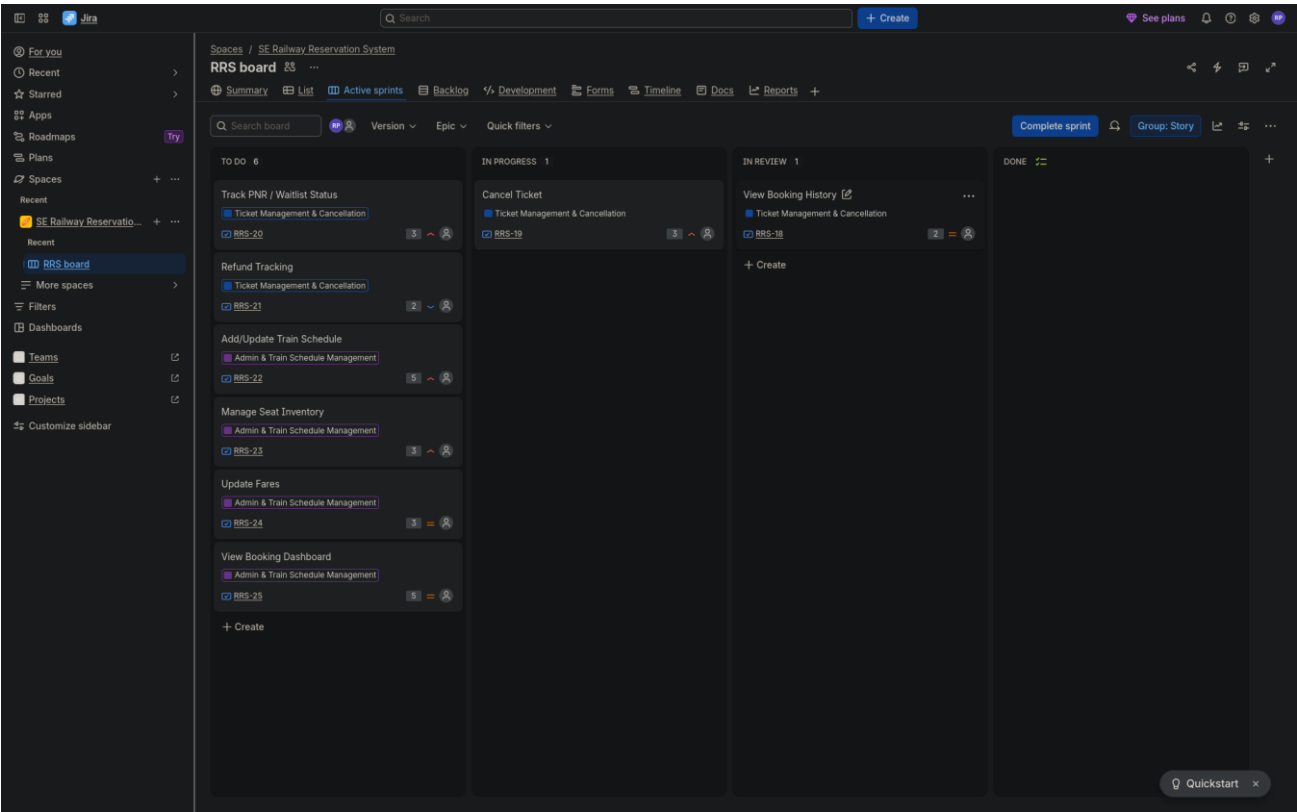
3.1 Backlog with Epics and User Stories



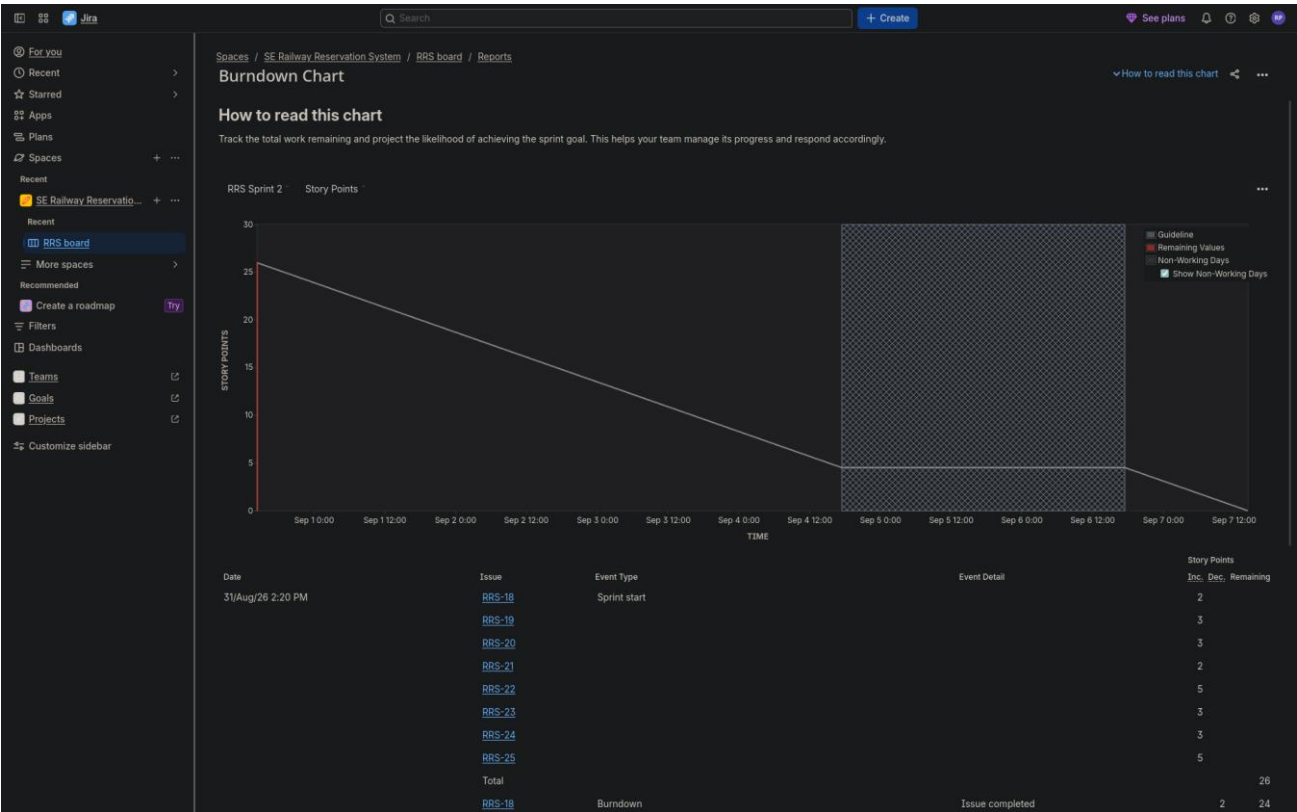
3.2 Story Point Assignments



3.3 Sprint Board (Active Sprint View)



3.4 Burndown Chart



4. Reflection Questions

1. Did your estimations reflect the actual effort?

For the most part, yes. Simpler stories such as Login and Profile Management (2 points) took noticeably less discussion time than higher-effort stories like Payment Processing (8 points), which involved more edge cases (failed transactions, multiple payment modes) and therefore more uncertainty. A few stories, such as Seat/Berth Selection, were slightly under-estimated at first because the dependency on real-time availability data wasn't fully accounted for until we broke the story down further.

2. Was your backlog well-prioritized?

The backlog was prioritized around the core booking flow first — Registration, Search, and Payment were marked High priority since they form the critical path a user must complete to book a ticket. Supporting features like Profile Management and Refund Tracking were marked Low/Medium since they enhance the experience but aren't blockers to a first working release. This ordering meant the first sprint focused on delivering an end-to-end bookable flow rather than isolated features.

3. How did your simulated sprint align with your plan?

Sprint 1 focused on the Registration & Authentication and Train Search epics, since booking cannot happen without an account and a way to find a train. Moving stories from To Do to In Progress to Done broadly matched the planned order, though Seat/Berth Selection took an extra cycle to move to Done because it depended on the search results being finalized first — a reminder that story sequencing inside a sprint matters, not just backlog priority.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed the remaining story points tracking close to the ideal guideline for the first half of the sprint, then flattening briefly around the Payment Processing story before dropping again as it was completed — consistent with it being the highest-effort (8-point) story in the sprint. This suggests the team's velocity is reasonably steady for smaller stories but that high-point stories should either be tackled earlier in the sprint or split further to avoid a late bottleneck.